HYDROCARBONS — FULL MIND MAP

Chemistry · Chapter 7.A · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

Topics in this chapter:

1. Organic Chemistry Basics & Origin-of-Life Elements

- Primitive-Earth simulation context mein source sequence Methane → Hydrogen cyanide → Nitrile → Amino acid diya gaya hai. Miller–Urey-type questions mein sequence ko option-wise recall karna useful hai. [Q1]
- Organic compounds aur bio-compounds ka backbone Carbon hai. Carbon atoms H, O, N, S, P jaise elements ke saath mainly covalent bonds bana kar enormous variety of molecules form karte hain. [Q2, Q3]
- Life-origin ke high-yield elements ko CHNOPS mnemonic se yaad rakho: Carbon, Hydrogen, Nitrogen, Oxygen, Phosphorus, Sulphur. Given-option Q4 mein Carbon + Hydrogen + Nitrogen wala set closest correct choice tha. [Q4]

2. Hydrocarbon Series, Alkene & Physical Properties

- Straight-chain alkane series mein molecular mass carbon count ke saath increase hota hai: Methane (CH_4) < Ethane (C_2H_6) < Propane (C_3H_8) < Butane (C_4H_{10}). [Q5]
- Do carbon atoms ke beech C=C double bond wala hydrocarbon Alkene kehlata hai. Open-chain mono-alkenes ka general formula C_nH_{2n} hota hai; Ethene, Propene aur Butene common examples hain. [Q6]
- Given compounds ka decreasing melting-point order source mein: Acetic acid > Chloroform > Ethanol > Methane. Ye trend specific substances ke actual melting points par based hai, sirf molecular mass par nahi. [Q7]
- Cigarette lighters mein commonly liquefied Butane (C_4H_{10}) fuel use hota hai. Butane flammable hydrocarbon/NGL hai jo pressure mein easily liquefy ho jata hai. [Q8]

3. Methane, Carbon Black, Biogas & Bhopal MIC

- Bhopal Gas Tragedy 2–3 December 1984 ki raat Union Carbide plant, Bhopal mein Methyl Isocyanate (MIC) leakage se hui. MIC ko methane ya methyl cyanide se confuse mat karo. [Q9, Q10, Q11]
- Methane colourless, odourless hydrocarbon hai jo marshes, anaerobic decomposition aur coal-related gases mein mil sakta hai. Methane/hydrocarbon ka incomplete combustion ya thermal decomposition Carbon black deta hai, jo black printing ink mein use hota hai. [Q12]
- Waterlogged paddy fields mein anaerobic microbial decomposition methane produce karti hai, isliye rice cultivation atmospheric methane ka important anthropogenic source hai. [Q13]
- Conceptual matching: Biogas → cow-dung/organic matter; Electrocardiography → heart disorder; DDT → insecticide; Nicotine → tobacco. [Q14]

⚠ EXAM TRAP — Q14 ka code-format source mein flawed/awkward hai

- ▸ Exam-safe conceptual matching relation yaad rakho; printed code sequence ko independent chemistry fact mat banao.
- ▸ Biogas ka major combustible component methane hota hai, lekin Q14 mein asked pairing “Biogas → Cow-dung” production-source ke context mein hai.

4. Fruit Ripening — Ethephon, Ethylene & Calcium Carbide

- Ethephon (Ethrel) plant-growth regulator hai jo plant tissues/aqueous conditions mein Ethylene release karke fruit ripening promote kar sakta hai. [Q15]
- Calcium carbide (CaC_2) moisture/water ke saath react karke Acetylene (C_2H_2) banata hai; acetylene ethylene-like ripening effect de sakta hai. [Q16, Q17]
- Natural fruit-ripening hormone Ethylene (C_2H_4) hai. Climacteric fruits mein ethylene ripening processes ko trigger/accelerate karta hai. [Q18]

⚠ EXAM TRAP — Artificial ripening mein Ethylene aur Calcium carbide ko same mat samjho

- ► Natural/physiological ripening agent = Ethylene; Ethephon ethylene release karta hai.
- ► Calcium carbide acetylene release karta hai aur food ripening ke liye unsafe/legally restricted practice hai; exam question source isse historical/artificial-ripening agent ke roop mein pooch sakta hai.

5. Acetylene — Welding, PVC & Preparation

- Gas welding mein Oxygen + Acetylene (oxy-acetylene) mixture high-temperature flame generate karta hai; steel welding/cutting ke classic questions isi pair par based hote hain. [Q19, Q20]
- Acetylene = Ethyne (C_2H_2). Welding ke alawa ye vinyl chloride/PVC-type chemical manufacture ka raw material route ho sakta hai. [Q21]
- Acetylene water + Calcium carbide (CaC_2) reaction se prepare kiya ja sakta hai; Silicon carbide + water statement incorrect hai. [Q21]

⚠ EXAM TRAP — Q21 — Calcium carbide vs Silicon carbide

- ► Correct reaction: $CaC_2 + 2H_2O \rightarrow C_2H_2 + Ca(OH)_2$.
- ► Silicon carbide (SiC, carborundum) ko acetylene-preparation reagent mat samjho.

6. Benzene Structure — Sigma & Pi Bonds

- Benzene ka formula C_6H_6 hai; six carbon atoms planar hexagonal ring form karte hain aur π-electrons ring mein delocalised hote hain. [Q22]
- Benzene mein total 12 sigma (σ) bonds hote hain: 6 C–C σ + 6 C–H σ. π system ko classical counting mein 3 π bonds ke equivalent count kiya jata hai. [Q22, Q23]

⚠ EXAM TRAP — Benzene mein alternating double-bond drawing ko literal fixed bonds mat samjho

- ► Exam counting: 12 σ + 3 π. Modern bonding picture mein π electrons poori ring par delocalised hote hain.
- ► Resonance ki wajah se benzene ke sab C–C bond lengths equivalent hote hain.

✓ Source-question mapping preserved: Q1–Q23